Step 1 – Prompt for AI Coding Assistant (Copilot)

Prompt:

Create a simple React functional component called GuestList that consumes a REST API endpoint using fetch().

Use the endpoint https://jsonplaceholder.typicode.com/users.

The component should:

- Fetch the guest data when the component loads.
- Display the guest names in a list.
- Show **"Loading guests..."** while the data is being fetched.
- Show an error message if the API request fails.
- Use React hooks such as useState and useEffect.
- Keep the code simple and beginner-friendly.
- Do not use any external libraries.

Step 2 – Component Generated

```
import { useEffect, useState } from 'react';


function GuestList() {
  const [guests, setGuests] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState('');


  useEffect(() => {
    async function fetchGuests() {
      try {
        const response = await fetch('https://jsonplaceholder.typicode.com/users');


        if (!response.ok) {
          throw new Error('Failed to load guests.');
```

```
      }

      const data = await response.json();
      setGuests(data);
    } catch (err) {
      setError(err.message || 'Something went wrong while loading guests.');
    } finally {
      setLoading(false);
    }
  }

  fetchGuests();
}, []);

if (loading) {
  return <p>Loading guests...</p>;
}

if (error) {
  return <p>{error}</p>;
}

return (
  <div>
    <h2>Guest List</h2>
    <ul>
      {guests.map((guest) => (
        <li key={guest.id}>{guest.name}</li>
```

```
      ))}
    </ul>
  </div>
 );
}

export default GuestList;
```

Step 2 – Add the Component

I created a React component called GuestList.jsx and added the code generated by Copilot.

The component uses:

- useState to store guests, loading status, and errors.
- useEffect to call the REST API when the component loads.
- fetch() to get the guest data.
- Loading and error states to handle different API conditions.

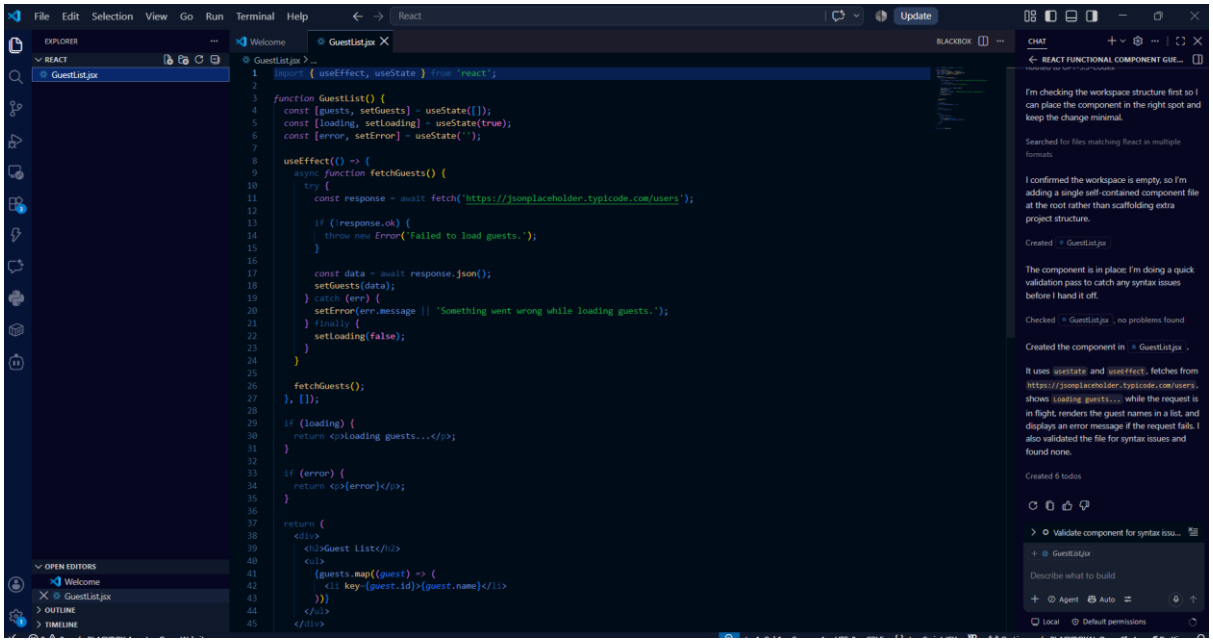
Step 3 – Display the Component

In App.jsx, import and use the component:

```
import GuestList from './GuestList';

function App() {
 return (
  <div>
   <GuestList />
  </div>
 );
}
```

export default App;



Step 4 – Test the REST API

I opened the application and verified that the component successfully fetched data from the REST endpoint.

The guest names were displayed under **Guest List**.

Expected result:

Guest List

Leanne Graham

Ervin Howell

Clementine Bauch

Patricia Lebsack

Chelsey Dietrich

...

Step 5 – Test Loading State

When the API request is being processed, the component displays:

Loading guests...

This is controlled by:

```
if (loading) {
  return <p>Loading guests...</p>;
}
```

Step 6 – Test Error State

To test the error handling, temporarily change the API URL to an invalid endpoint:

```
const response = await fetch(
  'https://jsonplaceholder.typicode.com/invalid'
);
```

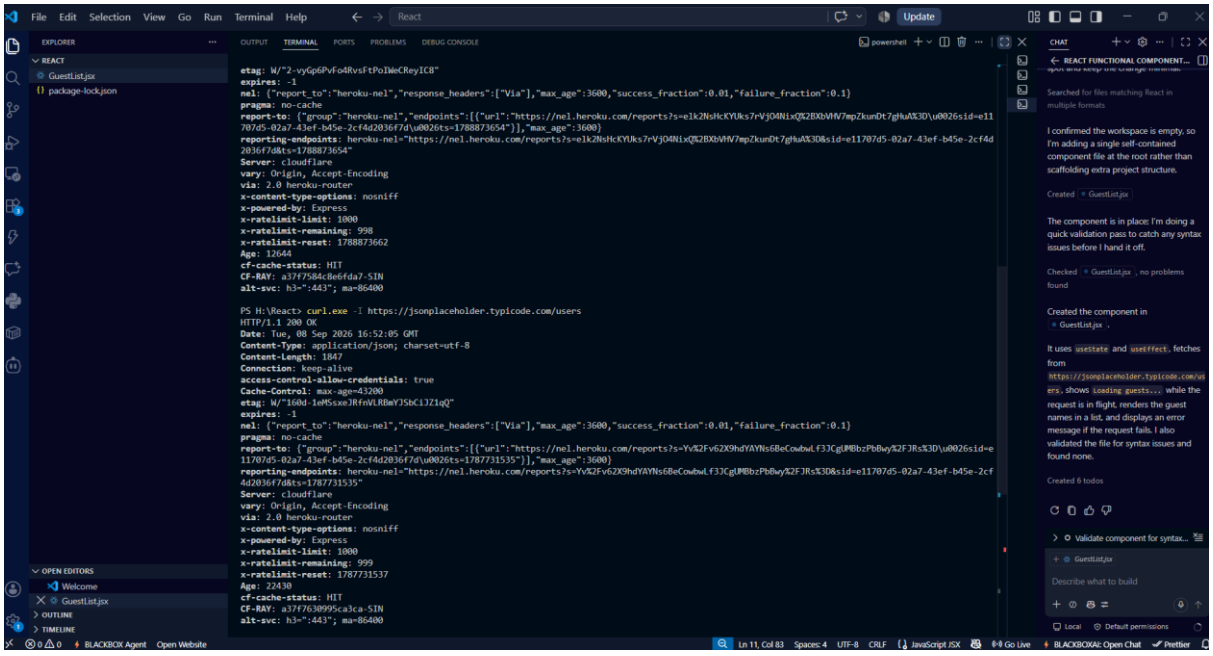
Run the application again.

The component should display:

Unable to load guests.

After testing, change the URL back to:

https://jsonplaceholder.typicode.com/users



Assignment 5

69336
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